FEMALE REPRODUCTIVE SYSTEM

- It consists of primary and accessory sex organs :
    - primary sex organs are a pair of ovaries
    - accessory sex organs are
        - Internal genitalia - fallopian tubes, uterus, cervix, vagina.
        - External genitalia - mons pubis, labia majora, labia minora, clitoris.

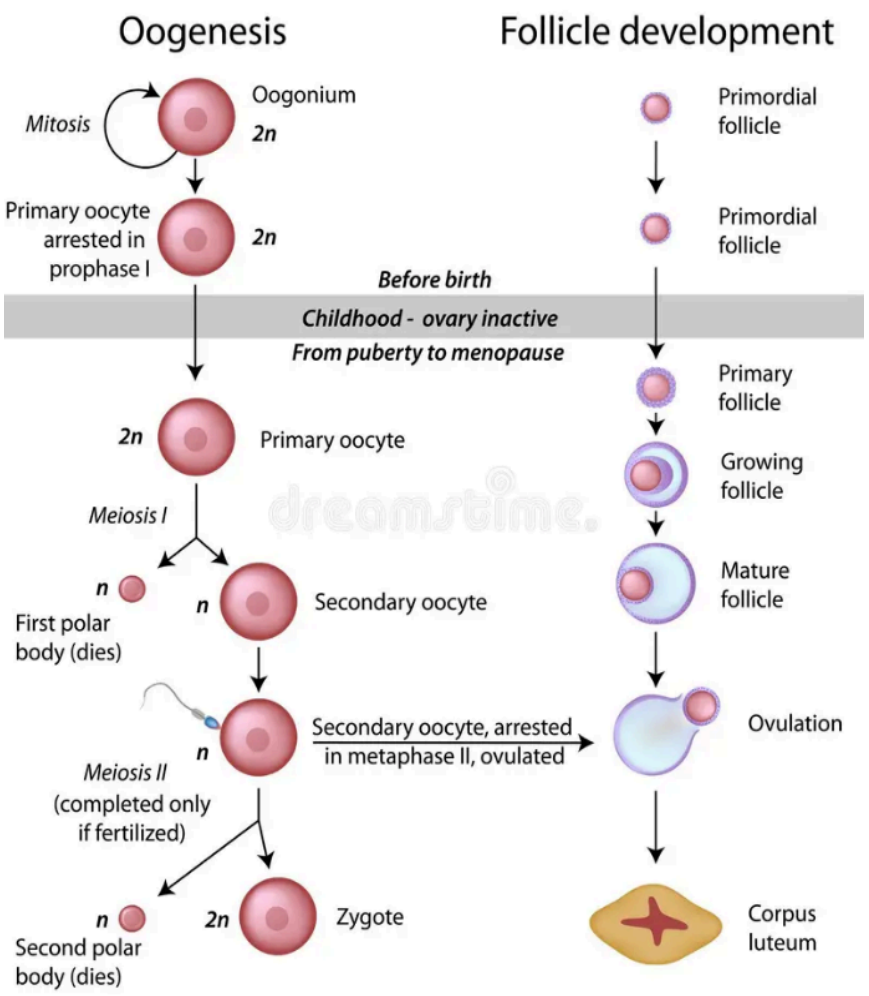
Functions of ovaries

- Gametogenic function - Oogenesis
- Endocrine function - secretion of ovarian hormones
- Menstrual cycle

OOGENESIS

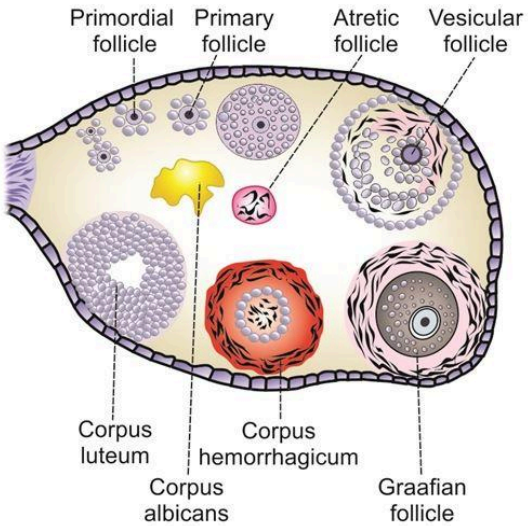
- It is the process of origin, development and maturation of ovum.
- Stages of oogenesis :
    - **Stage of proliferation** - in the foetal ovary, cells of germinal epithelium undergo repeated mitotic division and form diploid oogonia. No new oogonia are formed after birth.
    - **Stage of growth** - oogonia gives rise to diploid primary oocytes. primary oocytes enter a stage of prolonged prophase I of meiotic division and remain till puberty. the primary oocyte is enclosed in the primordial ovarian follicle.
    - **Stage of maturation** - after puberty, primary oocytes divide to form secondary oocyte and a first polar body. Secondary oocyte arrests in metaphase II of meiotic division. It contains a haploid number of chromosomes. First polar body is expelled.

After ovulation, the graafian follicle is ruptured and the secondary oocyte is released into the abdominal cavity. The meiosis II division of the secondary oocyte resumes only when the ovum (secondary oocyte) is fertilized. Here, the ovum is fused with sperm forming a diploid zygote along with the release of a second polar body.



<u>Ovarian follicles</u> - these are glandular structures made up of epithelial and granulosa cells that cover a developing immature ovum.

- Around the 8th week of intrauterine life, formation of primordial follicles begins.
- At the time of birth, about 1-2 million follicles are present in both ovaries.
- By the time of onset of puberty, due to atresia the number of follicles that develop reduces to about 3-4 lakhs.
- At the time of menarche, the follicles further reduced to 40-50 thousand, of which around 400-500 matured into an ova.



Different ovarian follicles are :

- Primordial follicle
- Primary follicle
- Vesicular follicle
- Graafian follicle

OVARIAN HORMONES

- Ovaries secretes female sex hormones such as :
    - estrogen
    - progesterone
    - inhibin
    - relaxin
    - ovarian androgens

Estrogen

- In a normal female (non pregnant) - estrogen is secreted
    - in large quantities by theca interna cells of ovarian follicles
    - in small quantities by corpus luteum of the ovaries.
- In a pregnant female, a large amount of estrogen is secreted by the placenta.

Functions

- Estrogen promotes the growth of ovarian follicles.
- It causes enlargement of the uterus and thickening of endometrium.
- It causes changes in fallopian tubes in order to facilitate the movement of ovum inside it.
- Also it brings changes in the cells lining the cervix and vagina.
- Estrogen is responsible for the development of secondary sexual characters in females such as :
    - hair distribution
    - softness and smoothness of skin
    - feminine body shape
    - fat deposition in breasts and buttocks
    - oval shape of pelvis
    - female high-pitched voice.
- Estrogen accelerates osteoblastic activity and bone growth.
- Estrogen induces anabolism of proteins.
- Estrogen causes fat deposition in the subcutaneous tissues.
- Estrogen causes sodium and water retention from the renal tubules.

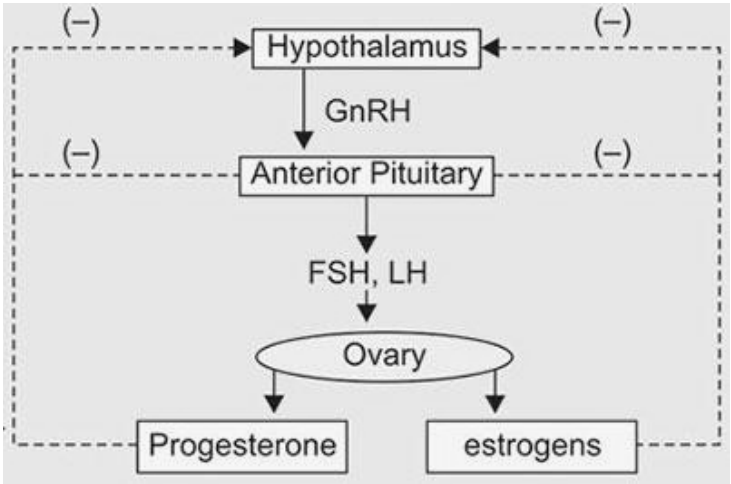
Progesterone

- Progesterone is secreted in :
    - a small quantity by theca interna cells during follicular phase.
    - a large quantity by corpus luteum during secretory phase.
- During pregnancy,
    - a large amount is secreted by corpus luteum (Ist trimester).
    - in the 2nd and 3rd trimesters the action is taken over by placenta.

Functions

- Progesterone promotes the secretory activities of uterine endometrium and fallopian tubes.
- It prepares the uterus for the implantation of a fertilized ovum.
- It increases the thickness of cervical secretions and vaginal epithelium.
- It promotes development of the lobules and alveoli of mammary glands.
- Progesterone increases the basal body temperature after ovulation.
- It increases the ventilation via the respiratory centre.

Regulation of ovarian hormones



- Hypothalamus secrete gonadotropin releasing hormone (GnRH)
- Anterior pituitary releases follicle stimulating hormone (FSH) and luteinizing hormone (LH)

- FSH stimulates the theca and granulosa cells to secrete Estrogen.
- LH activates the corpus luteum to secrete Progesterone.
- Estrogen inhibits secretion of FSH and GnRH by negative feedback.
- Progesterone inhibits release of LH by negative feedback mechanism.

MENSTRUAL CYCLE

- It is defined as the cyclic events that take place in a rhythmic fashion in the primary and accessory sex organs during the reproductive period of a woman's life.

- Normal average duration of a menstrual cycle is 28 days.
- Menarche **:** commencement of first menstrual cycle at the age of 12-15 years.
- Menopause**:**permanent stoppage of menstrual cycle at the age of 45-50years

- It comprises changes taking place in different reproductive structures like :
    - Ovarian changes (ovarian cycle)
    - Uterine**/**endometrial changes (uterine**/**endometrial cycle)
    - Changes in cervix
    - Changes in vagina

OVARIAN CYCLE

- It refers to the cyclic changes occurring in ovaries during a menstrual cycle.
- It is explained in three phases :
    - **Follicular phase (pre-ovulatory phase)**
        - in this phase the development of ovarian follicles and maturation of ovum occurs.
        - it extends from the 5th - 14th day of the menstrual cycle.
        - Primordial follicles start maturing and form into primary follicles which are then developed into vesicular follicles and matures into a graafian follicle with secondary oocyte (ovum) inside it.
        - Follicular phase occurs under the influence of FSH and results in the secretion of Estrogen by follicles.

- **Ovulation**
  - It is the process by which a mature graafian follicle is ruptured and ovum (secondary oocyte) is released into the abdominal cavity.
  - It occurs on the 14th day of the menstrual cycle.
  - LH surge (secretion of large amount of luteinizing hormone prior to ovulation) results in the process of ovulation.
  - Only a single mature ovum is released from any one of the ovaries during each menstrual cycle.
  - Ovum is viable only for 12 to 24 hours
- **Luteal phase (post-ovulatory phase)**
  - After ovulation, the ruptured graafian follicle fills with blood to form corpus hemorragicum.
  - It is then transformed into a glandular yellow body called corpus luteum which is the reason for luteal phase.
  - It extends between the 15th to 28th day of the menstrual cycle.
  - corpus luteum is developed under the influence of LH and as a result progesterone is secreted.

If the ovum is fertilized, the corpus luteum of pregnancy persists for about 3months to produce progesterone and estrogen ; afterwards the action is taken by placenta.
If the ovum is not fertilized, corpus luteum degenerates into corpus luteum menstrualis and is transformed into a whitish scar called corpus albicans.

UTERINE / ENDOMETRIAL CYCLE

- It refers to the cyclic changes that occur in the endometrial lining of the uterus.
- It takes place in three phases :
  - **Menstrual phase / bleeding phase**
    - It is the process of desquamation and expel of endometrium along with blood and fluid through vagina.
    - It extends from the 1st to 5th day of the menstrual cycle.
    - Sharp decline in the plasma levels of Estrogen and Progesterone are responsible for the menstrual bleeding.
    - At the end of menstruation, the thickness of the endometrium is about 1mm.
  - **Proliferative phase**
    - It is the phase of proliferation of endometrial cells and development of uterine glands along with blood vessels.
    - It extends from 6th to 14th day of the menstrual cycle.
    - It corresponds to the follicular phase of the ovarian cycle.
    - Proliferative phase occur under the influence of Estrogen secreted by ovarian follicles.
    - At the end of the proliferative phase, the thickness of endometrium is about 3 to 4mm.
  - **Secretory phase**
    - It is the preparatory period prior to the implantation of ovum, if the fertilization has occurred.
    - It extends between the 15th to 28th day of the menstrual cycle.
    - It corresponds to the luteal phase of the ovarian cycle.

- During this phase, uterine glands and blood vessels become more coiled inorder to supply the thickened endometrium.
- This phase occurs under the influence of increased secretion of Estrogen and Progesterone by ovarian follicles.
- At the end of the secretory phase, the thickness of the endometrium is increased up to 6mm.

If the ovum is fertilized and implanted in the uterus, the bleeding phase of the next uterine cycle is abolished and pregnancy continues.

If the ovum is not fertilized and implanted, the phase of menstruation occurs and a new uterine cycle continues.

HORMONAL REGULATION OF MENSTRUAL CYCLE

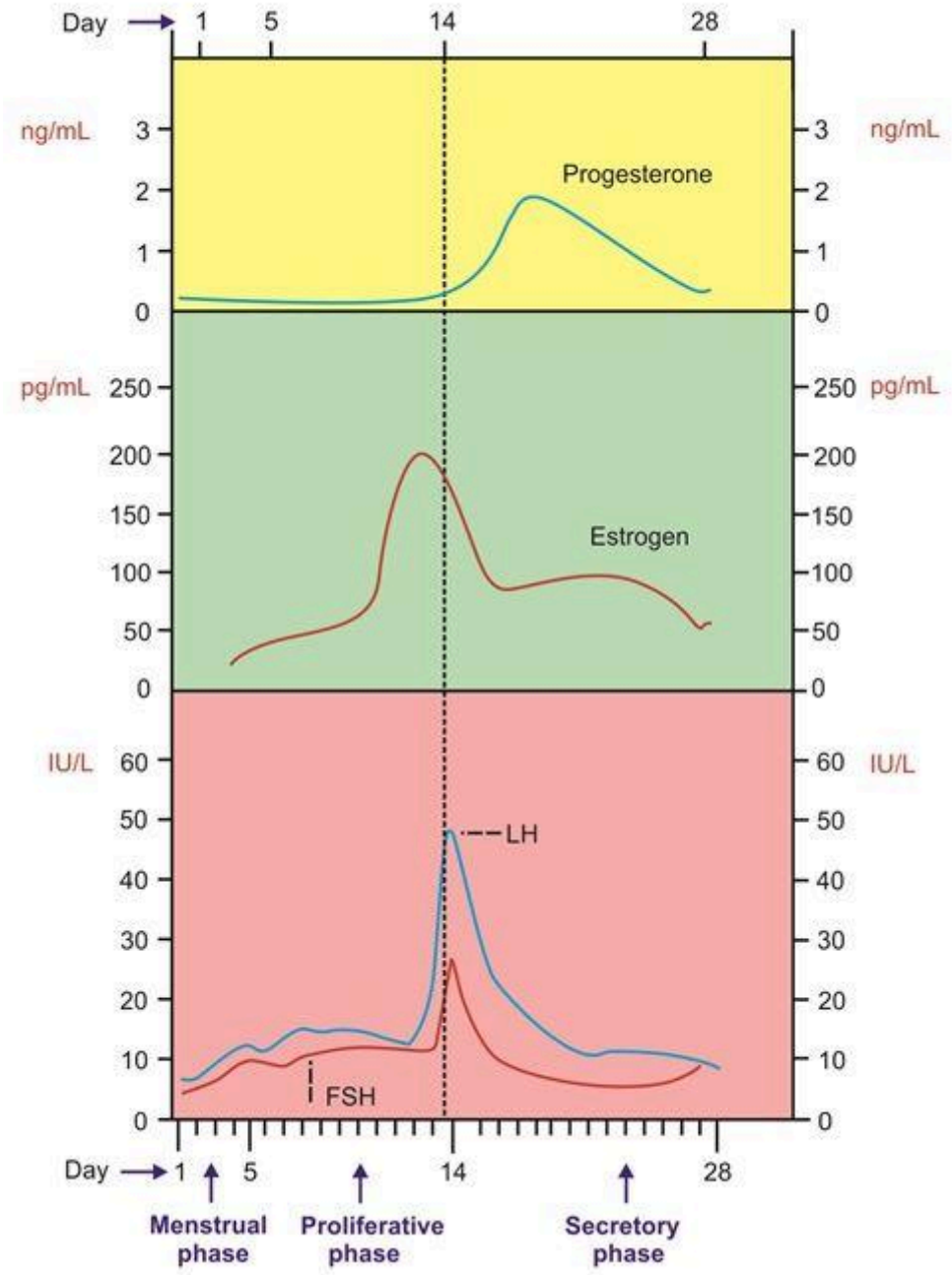
- Menstrual cycle is regulated by hypothalamic-hypophyseal-ovarian axis.
- Hypothalamic hormone : gonadotropin releasing hormone(GnRH)
    - It is responsible for the onset of the menstrual cycle.
    - It stimulates anterior pituitary and secretion of FSH and LH.
- Anterior pituitary hormones : FSH and LH
    - FSH stimulates the growth of ovarian follicles
    - LH triggers ovulation and formation of corpus luteum.
- Ovarian hormones :Estrogen and Progesterone
    - Ovarian follicle secretes large quantity of Estrogen
    - Corpus luteum secretes large quantity of Progesterone.

In ovarian cycle :

- Estrogen secretion is increased during follicular and luteal phase.
- Progesterone secretion is increased during luteal phase and is almost absent during follicular phase.

In uterine cycle :

- Estrogen is responsible for the proliferative and secretory phase.
- Progesterone is responsible for the secretory phase and is almost absent in the proliferative phase.



CONTRACEPTIVE METHODS

- It is the fertility control method or device that is used to prevent pregnancy.
    - Rhythm method
    - Barrier methods
    - Hormonal contraceptives
    - Intrauterine contraceptive device
    - Medical termination of pregnancy
    - Surgical method

Rhythm method

- It is based on the time of ovulation. Also known as a natural method.
- Pregnancy is prevented by avoiding sexual intercourse during the fertile/dangerous period (3-4 days before and 3-4 days after ovulation) is called the rhythm method.
- The days when pregnancy does not occur are called as infertile/safe period (4-5 days after menstrual bleeding and 5-6 days before the onset of the next menstrual cycle).
- Because ovum is viable only for 12 to 24 or 48 hours and sperms are viable for 24 to 48 or 72 hours.

Advantages

- It is one of the most successful fertility control methods in women with regular periods and knows the exact day of ovulation.
- It has no physical side effects.

Disadvantages

- It can be difficult for the women with irregular periods, as they cannot predict the exact day of ovulation.
- It does not give any protection against sexually transmitted infections.
- This method is not reliable.

In males : withdrawal of penis from vagina before ejaculation during sexual intercourse.

Barrier methods

- Different barrier methods are used to prevent the entry of sperm into the uterine cavity during sexual intercourse.
- They are of two types :
    - **Physical barriers**
        - Condoms
            - prevent the entry of semen into the female genital tract.
            - most commonly used barrier by men.

              Advantages
                - gives protection against sexually transmitted infections(STI) including HIV.
                - most effective contraceptive device if used properly.
                - safe and easily available.

              Disadvantages
                - chances of tear or slip off, if not used correctly.
                - may decrease sexual sensation.
                - they are single time use and throw methods.
        - Cervical cap
            - does not contain hormones,but used with spermicidal gel.
            - not an effective contraceptive device
            - does not protect against STI.
        - Diaphragm
            - safest device.
            - does not protect the spread of STI
            - cause irritation in vagina and risk of infections.
        - Vaginal sponge
            - effective device when used correctly.
            - does not protect against STI.
    - **Chemical barriers** - spermicidal substances (cream,jelly,paste) are inserted into vagina before sexual intercourse. It destroys the sperms.
            - cheap and easy to use
            - spermicides cause vaginal irritation.

Hormonal contraceptives

- Contraceptives that prevent pregnancy by inhibiting maturation of follicles and ovulation.

Oral contraceptives

- These are the drugs taken orally (as pills) to prevent pregnancy.
- These pills contain synthetic estrogen and synthetic progesterone.

Contraceptive pills are of four types :

- **Combined pills**
    - contain a moderate dose of synthetic estrogen and a mild dose of synthetic progesterone.
    - a relatively large amount of these hormones in blood suppresses the release of FSH and LH by negative feedback mechanism, thereby preventing the maturation of follicles and ovulation.
    - pills are taken daily from the 5th to 25th day of the menstrual cycle.
- **Sequential pills**
    - contain a high dose of estrogen and moderate dose of progesterone.
    - pills are taken daily from 5th to 20th day and then from 23rd to 28th day.
- **Mini pills**
    - contain a low dose of only progesterone.
    - increases thickness of cervical mucosa, inhibits sperm motility.
    - prevents pregnancy but does not affect ovulation.
    - taken throughout the menstrual cycle.
- **Emergency contraceptive pills**
    - contain either synthetic progesterone or synthetic estrogen.
    - are taken to prevent pregnancy after unprotected sex.

Advantages

- are administered in correcting irregular menstruations.
- most effective at preventing pregnancy if taken correctly.

Disadvantages

- long term use of contraceptive pills may cause side effects :
    - Hypertension, Heart attack, stroke and cancer.

Intrauterine contraceptive device (IUCD)

- It is a device that is inserted into the uterus inorder to prevent fertilization and implantation of ovum.
- Three types of IUCD :
    - **Non medicated IUCD**
        - a common one is the Lippes loop which is S shaped and made from plastic.
        - causes a mild and sterile inflammatory reaction in the uterine wall that initiates spermicidal action.
        - First generation IUCD which is outdated now.

    - **Copper IUCD**
        - a common one is the Copper T which is T shaped and made from plastic and copper.
        - it releases free copper and copper salts that cause mild changes in endometrial and cervical secretions to prevent entry of sperm.
        - Second generation IUCD

    - **Hormone-releasing IUCD**
        - Third generation IUCD. a common one is progestasert.
        - it releases progesterone slowly into the uterus after its insertion.
        - prevents sperm entry and implantation of ovum.

Advantages

- provide long term-protection (5-10 years) against pregnancy.
- 99% effectiveness at preventing pregnancy.
- low maintenance and works immediately after insertion.

Disadvantages

- rarely IUCDs slips out of place.
- does not protect against STDs.
- cause heavy bleeding or longer periods.

Medical termination of pregnancy (MTP)

- Also known as abortion which is done during first few months of pregnancy.
- Three ways of MTP :
    - **Dilatation and curettage (D and C)**
        - The cervix is dilated and the implanted zygote is desquamated.
    - **Vacuum aspiration**
        - Removal of implanted zygote up to 12 weeks of pregnancy.
    - **Administration of prostaglandin**
        - Expulsion of implanted zygote by administration of prostaglandins and inducing uterine contractions.

Surgical / permanent method (sterilization)

- Permanent sterility is obtained by surgical methods.

In females : Tubectomy

- Fallopian tubes are cut and both ends are ligated (tied).
- Prevents the entry of ovum into the uterus.
- If necessary, recanalization of the fallopian tube can be done.

In males : Vasectomy

- Vas deferens is cut and the ends are ligated.
- Prevents the entry of sperms into the ejaculatory duct.
- Semen is devoid of sperms.
- If necessary, recanalization of vas deferens can be done.

FERTILIZATION

- It is the fusion of male and female gametes (sperm and ovum) to produce new offsprings
- It takes place in the fallopian tube.
    - During ovulation, ovum is released into the abdominal cavity.
    - If sexual intercourse occurs at this time and semen is ejaculated in the female genital tract, the sperms enter into the vagina and uterus to reach the fallopian tube.
    - Among 200-300 million of sperms only one sperm fertilizes the ovum.
    - The sperm penetrates the outer layers of ovum with the help of hyaluronidase and proteolytic enzymes present in the acrosome of sperm.
    - By this the genetic material of sperm enters the oocyte causing fertilization and formation of zygote.

IMPLANTATION

- It is the process by which the fertilized ovum implants (gets attached) in the endometrial wall of the uterus.
- The fertilized ovum starts dividing immediately and forms morula and blastocyst (different cell stages).
- During this, the uterus is already prepared and is in the secretory phase by the action of progesterone.
- After implantation and different cell stages the zygote develops into an embryo inside the womb.
- The endometrium after implantation is called a decidua. The decidual cells are the source of nutrition for the embryo till the placenta takes up this function.

PLACENTA

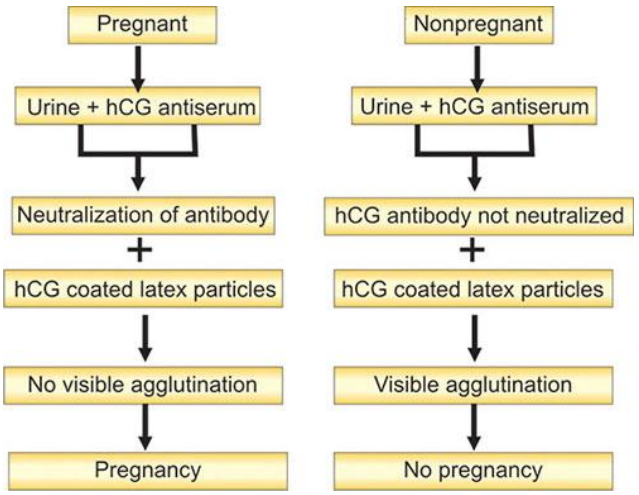
- It is a temporary membranous vascular organ that develops during pregnancy and forms a link between the foetus and mother.
- It is implanted in the wall of the uterus.
- By the 12th week of pregnancy the placenta becomes fully functional.

Functions of placenta

- Endocrine function - it secretes a number of hormones :
    - Human chorionic gonadotropin(hCG)
    - Human chorionic somatomammotropin(HCS)
    - Human chorionic thyrotropin(HCT)
    - Placental Estrogen
    - Placental Progesterone
    - Relaxin
- Respiratory function - since the fetal lungs are non functioning, placenta helps for the exchange of gases between fetal and maternal circulations.
- Nutritive function - the substances necessary for the development of the fetus diffuse from maternal to fetal circulation through placenta.
- Excretory function - the metabolic end products and waste materials from the fetal body are excreted into mothers circulation through placenta.
- Protective function - is because not all the antibodies from maternal circulation can get passed through the placenta to reach the fetal body.

PREGNANCY TEST

- It is the immunological test used to detect or confirm pregnancy.
- It is based on the presence or absence of hCG in the urine of a woman.



- hCG antiserum contains antibodies against hCG.
- If the urine contains hCG, absence of agglutination of latex particles - woman is pregnant.
- If the urine doesn't contain hCG, presence of agglutination of latex particles - non pregnant woman.

MAMMARY GLANDS

- These are one of the important glands of the female reproductive system.
- The growth and enlargement of mammary glands begins during puberty and reaches its maximum during pregnancy.
- It is under the influence of several hormones like, estrogen, progesterone, prolactin, oxytocin.
- The major functions of breasts are found in the mother-hood of a female :
    - Synthesis of breast milk
    - Storage of breast milk
    - Secretion of breast milk
    - Mammary glands provide nourishment to newborn through breastfeeding.

LACTATION

- It is the process of synthesis, secretion and ejection of milk.
- It involves two process :
    - **Milk secretion** - it refers to synthesis of milk by alveolar epithelium and its passage through the duct system. It occurs in two phases :
        - **Lactogenesis** - it is the process of initiation of milk secretion. The milk secreted initially is called colostrum. The hormone prolactin promotes lactogenesis.
        - **Galactopoiesis** - it is the process of maintenance of milk secretion. It depends on prolactin, growth hormone, thyroxine and cortisol.
    - **Milk ejection** - it is the process of discharge of milk from mammary glands. It is a reflex phenomenon that depends upon the suckling effect by the baby. It is also known as milk ejection reflex or milk let down reflex. It is under the influence of the hormone oxytocin.